



**INTRODUCTORY PHYSICS LABORATORY
DEPARTMENT OF PHYSICS
FACULTY OF SCIENCE AND DATA ANALYTICS
INSTITUT TEKNOLOGI SEPULUH NOPEMBER
SURABAYA**



NAME :
STUDENT ID :
ASSISTANT :
DATE :

EXPERIMENT: FLETCHER’S TROLLEY

Objectives:

1. To calculate the velocity and acceleration in a fletcher's trolley system.
2. To analyze the graph of the relationship between distance and time.

EXPERIMENT SUMMARY

INTRODUCTION

Motion is the change in position of an object relative to a reference point over time. In kinematics, motion is described by quantities such as displacement, velocity, and acceleration without considering the causes of the motion. Understanding the concept of motion is fundamental to studying various mechanical phenomena, as almost all mechanical events involve changes in an object's position over time. To experimentally study the characteristics of motion, an apparatus capable of producing measurable motion and facilitating observation is required. One of the most widely used tools in physics laboratories is the Fletcher's Trolley. This apparatus is designed to produce motion of an object along a straight track, allowing observation and measurement of kinematic quantities to be conducted in a more controlled manner during the experiment [1].

The Fletcher's Trolley is a practical apparatus used to observe the motion of an object along a straight track. This apparatus consists of a trolley that moves on a track and is connected to a load by a string that passes over a pulley. In the experiment, the track is assumed to be frictionless so that the frictional force between the trolley and the track can be neglected, while the pulley is considered ideal so that its mass and friction do not affect the motion of the system. When the load is released, the trolley moves along the track with a relatively constant acceleration. This condition allows the motion of the trolley to be approximated as uniformly accelerated linear motion (UALM). Thus, the Fletcher's Trolley enables the measurement of kinematic quantities, namely distance, time, and acceleration, during the motion [4].

An object is said to be in motion if its position changes relative to a reference point over time. One of the simplest forms of motion is uniform linear motion (ULM), which is the motion of an object along a straight path with constant velocity. The magnitude of velocity (v) in uniform linear motion is expressed by the following equation.

$$v = \frac{s}{t} \quad (1)$$

where v is the velocity in meters per second (m/s), s is the distance traveled in meters (m), and t is the time in seconds (s). This relationship shows that velocity is obtained from the ratio between the distance traveled and the time required by the object to cover that distance [2].

If the velocity of an object changes over time, the object undergoes uniformly accelerated linear motion (UALM) if its acceleration is constant. The relationship between distance, time, and acceleration in uniformly accelerated linear motion is expressed by the following equation.

$$s = v_0 \cdot t + \frac{1}{2} \cdot a \cdot t^2 \quad (2)$$

since the object starts moving from rest, the initial velocity is zero ($v_0 = 0$), so the equation becomes.

$$s = \frac{1}{2} \cdot a \cdot t^2 \quad (3)$$

$$a = \frac{2s}{t^2} \quad (4)$$

where a is the acceleration in meters per second squared (m/s^2), s is the distance traveled in meters (m), and t is the time in seconds (s). The relationship between distance, time, and acceleration shows that the magnitude of acceleration can be determined from measurements taken during the motion. The acceleration value obtained is used to analyze whether the motion of the object satisfies the characteristics of uniformly accelerated linear motion, namely having constant acceleration [3].

METHODOLOGY

A. Tools and Materials

Listed below are the tools and materials used in the experiment:

Table 1. Tools and materials for the fletcher’s trolley experiment.

Tools	Quantity	Function
Fletcher’s trolley	1 set	Used to conduct the fletcher's trolley experiment.
Loads	1 set	Used to provide the pulling force on the trolley.
Tape measure	1 piece	Used to measure the distance traveled by the trolley
String	1 piece	Used to connect the trolley to the loads.
Pulley	1 piece	Used to change the direction of the pulling force.
Digital scale	1 piece	Used to measure the mass of the loads.
Stopwatch	1 piece	Used to measure the time taken by the trolley to travel along the track.

B. Procedure

The steps for this experiment are as follows.

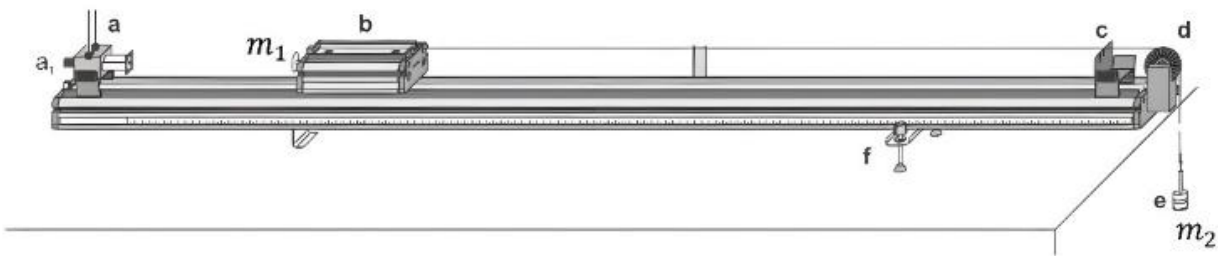


Figure 1. Experimental setup of the fletcher’s trolley.

1. Set up the experimental setup as shown in Figure 1.
2. Measure the mass of the trolley as m_1 and the mass of the hanging load as m_2 , then record the measurement results after confirmation by the laboratory assistant.
3. Place the trolley on the track and set the distance (s) along the track using a tape measure according to the given variation of 50 cm, 60 cm, 70 cm, dan 80 cm. Ensure that the system is initially at rest.
4. Hang the load m_2 on the end of the string that passes over the pulley. Ensure that the string is taut and the load is not oscillating, then release the system so that the trolley begins to move.
5. Measure the time (t) required for the trolley to travel the distance (s) using a stopwatch.
6. Record the distance (s) and time (t) data obtained.
7. Repeat steps 3 – 6 ten times as repetitions for each distance variation with the same hanging load m_2 .
8. Repeat steps 2 – 7 with different hanging load variations where $m_2 > 20\text{ g}$ for each distance variation that has been determined.

C. Experiment Flowchart

DATA ANALYSIS AND DISCUSSIONS

A. Data Analysis

The following are the tables of measurement result obtained in the laboratory:

Table 2. Measurement results for time at a load mass variation of g at a distance of 50 cm.

Repetition	m_1 (g)	m_2 (g)	s (cm)	t (s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Table 3. Measurement results for time at a load mass variation of g at a distance of 60 cm.

Repetition	m_1 (g)	m_2 (g)	s (cm)	t (s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Table 4. Measurement results for time at a load mass variation of g at a distance of 70 cm.

Repetition	m_1 (g)	m_2 (g)	s (cm)	t (s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Table 5. Measurement results for time at a load mass variation of g at a distance of 80 cm.

Repetition	m_1 (g)	m_2 (g)	s (cm)	t (s)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				

Approved by:

Laboratory Assistant

B. Calculation Analysis

Calculate the velocity of the trolley just before it hits the stopper at point C (see Figure 1) and the acceleration experienced by the trolley for all distance variations using Equations (1) and (4)!
The following is a sample calculation based on the measurement data.
Sample calculation:

The following is the table of calculation results obtained:

Table 6. Calculation results for velocity and acceleration at a load mass variation of g at a distance of 50 cm.

Repetition	m_1 (kg)	m_2 (kg)	s (m)	t (s)	v (m/s)	a (m/s ²)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
Average						
Standard Deviation						
Uncertainty						

Table 7. Calculation results for velocity and acceleration at a load mass variation of g at a distance of 60 cm.

Repetition	m_1 (kg)	m_2 (kg)	s (m)	t (s)	v (m/s)	a (m/s ²)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
Average						
Standard Deviation						
Uncertainty						

Tabel 8. Calculation results for velocity and acceleration at a load mass variation of g at a distance of 70 cm.

Repetition	m_1 (kg)	m_2 (kg)	s (m)	t (s)	v (m/s)	a (m/s ²)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
Average						
Standard Deviation						
Uncertainty						

Tabel 9. Calculation results for velocity and acceleration at a load mass variation of g at a distance of 80 cm.

Repetition	m_1 (kg)	m_2 (kg)	s (m)	t (s)	v (m/s)	a (m/s ²)
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
Average						
Standard Deviation						
Uncertainty						

C. Graph

Based on the data processing results obtained, plot the graph of distance (s) versus time squared (t^2) for each distance variation! Determine the linear regression equation from the graph!

D. Discussion

CONCLUSION

From this experiment, it can be concluded that:

1.

2.

REFERENCES

- [1] M. Abdullah, *Fisika Dasar I*, Edisi Revisi. Bandung, Indonesia: Penerbit ITB, 2016.
- [2] D. C. Giancoli, *Physics: Principles with Applications*, 7th ed. Upper Saddle River, NJ, USA: Pearson Education, 2014.
- [3] D. Halliday, R. Resnick, and K. S. Krane, *Fundamentals of Physics*, 8th ed. Hoboken, NJ, USA: John Wiley & Sons, 2010.
- [4] D. Halliday, R. Resnick, and J. Walker, *Fundamentals of Physics Extended*, 9th ed. Hoboken, NJ, USA: John Wiley & Sons, 2010.
- [5] R. A. Serway and J. W. Jewett, *Physics for Scientists and Engineers with Modern Physics*, 9th ed. Mason, OH, USA: Brooks/Cole, Cengage Learning, 2012.

PRELIMINARY ASSIGNMENT

Answer the following questions before conducting the experiment.

1. What is meant by uniformly accelerated linear motion (UALM)?
2. Derive the formula for the acceleration of the trolley (a) if the only known values are distance (s) and time (t) using the UALM equation!
3. Explain the relationship between distance (s) and time (t) based on the UALM equation!
4. Why does this experiment use a trolley that moves on a smooth track rather than a rough surface? Explain its relation to the concept of UALM!
5. If the hanging load is increased, explain the effect on the acceleration value and the slope of the graph physically!